

2025

MATHEMATICS — MINOR

Paper : MN-3

(Ordinary Differential Equations and Group Theory)

Full Marks : 75

*The figures in the margin indicate full marks.**Candidates are required to give their answers in their own words as far as practicable.**All the notations and symbols have their usual meanings.*

Group - A

(Ordinary Differential Equations)

Answer *any nine* questions.

1. (a) Solve : $\sqrt{1-x^2} dy + \sqrt{1-y^2} dx = 0$.

(b) Show that the general solution of the equation $\frac{dy}{dx} + Py = Q$ can be written as $y = K(u - v) + v$,
where K is a constant and u, v are its two particular solutions. 2+3

2. Define Integrating factor of a differential equation and solve : $(x^2y - 2xy^2)dx - (x^3 - 3x^2y)dy = 0$. 1+4
Non exact

3. Solve : $x \frac{dy}{dx} + y = y^2 \log x$. 5
IF = $\frac{1}{x^2 \log x}$

4. Reduce the differential equation $x^2 p^2 + py(2x + y) + y^2 = 0$ to Clairaut's form by the substitution $y = u$,
 $xv = v$ and solve it to find general solution and singular solution. 1+2+2

5. Solve $(D^2 + 1)y = 3\cos^2 x$, where $D \equiv \frac{d}{dx}$ by the method of D -operator. 5

6. Solve by the method of variation of parameters : $\frac{d^2 y}{dx^2} + 9y = \sec 3x$. 5

7. Solve by the method of undetermined coefficients $\frac{d^2 y}{dx^2} - 7 \frac{dy}{dx} + 6y = (x - 2)e^x$. 5

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8. Solve and find the singular solution of $x^2 p^2 - 2py + 4x = 0$ $\left(p = \frac{dy}{dx}\right)$. 5

9. Solve $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = 0$, given that $\left(x + \frac{1}{x}\right)$ is one integral (or solution). 5

10. Solve the following Simultaneous Linear differential equations : 5

$$(D - 1)x + Dy = 2t + 1$$

$$(2D + 1)x + 2Dy = t, \text{ where } D \equiv \frac{d}{dt}.$$

11. Show that the family of confocal conics $\frac{x^2}{(a^2 + \lambda)} + \frac{y^2}{(b^2 + \lambda)} = 1$ is self-orthogonal, λ is a parameter. 5

12. Solve : $\frac{d^2 y}{dx^2} + 2 \sin x \frac{dy}{dx} + 2y \cos x = 0$. 5

13. Solve the equation : $(2x^2 y - 3y^4)dx + (3x^3 + 2xy^3)dy = 0$. 5

14. Solve the simultaneous equation : $\frac{x dx}{z^2 - 2yz - y^2} = \frac{dy}{y + z} = \frac{dz}{y - z}$. 5

15. Solve the differential equation : $(D^3 - 5D^2 + 7D - 3)y = e^{2x} \cosh x \left[D \equiv \frac{d}{dx}\right]$. 5

16. Find an integrating factor of the differential equation $(y^2 + 2x^2 y)dx + (2x^3 - xy)dy = 0$ and hence solve it. 2+3

$$y(y + 2x^2) + x(2x^3 - y)dy = 0$$

Group - B

(Group Theory)

Answer *any six* questions.

17. Prove that the set Q of all rational numbers other than 1 under the binary operation $*$ defined by $a * b = a + b - ab$ forms an Abelian group for all $a, b \in Q - \{1\}$. 5

18. Define order of a group. Also, explain the order of an element in a group and hence prove that each element of a finite group is of finite order. 1+1+3

19. Let G be a group and H, K are two subgroups of G . Prove that $H \cap K$ is also a subgroup of G . Is $H \cup K$ also a subgroup? Justify your answer. 3+1+1

20. Let $G = \left\{ \begin{bmatrix} a & 0 \\ b & 1 \end{bmatrix} : a, b \in \mathbb{R}, a \neq 0 \right\}$. Show that G becomes a group under usual matrix multiplication. 5

21. If $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 6 & 4 & 7 & 5 & 2 & 3 & 1 \end{pmatrix}$ be an element of S_7 , then find α^{-1} . Is it an even permutation? 3+2

22. Define subgroup of a group G . Let G be a multiplicative group and H be its subset defined by :
 $H = \{x : x \in G \text{ \& } xa = ax, \forall a \in G\}$.

Prove that H is a subgroup of G .

1+4

23. Prove that every group of prime order is cyclic. 5

24. Show that the 8th roots of unity forms a cyclic group. 5

25. Prove that every cyclic group is a commutative group. Is the converse true? Justify your answer. 3+1+1

26. Prove that for a cyclic group G , there exists an element a in G such that $o(a) = o(G)$. 5

$$2y + 2n^2$$

$$6n^2 = y$$

$$4n^2 - 3y$$

$$n^2 + 2n^3 - 2n^3 + n^2$$

$$\frac{1}{2n^2}$$

$$\frac{2n^2}{2n^2} = 1$$

$$\frac{-n^2}{2n^2} = -\frac{1}{2}$$